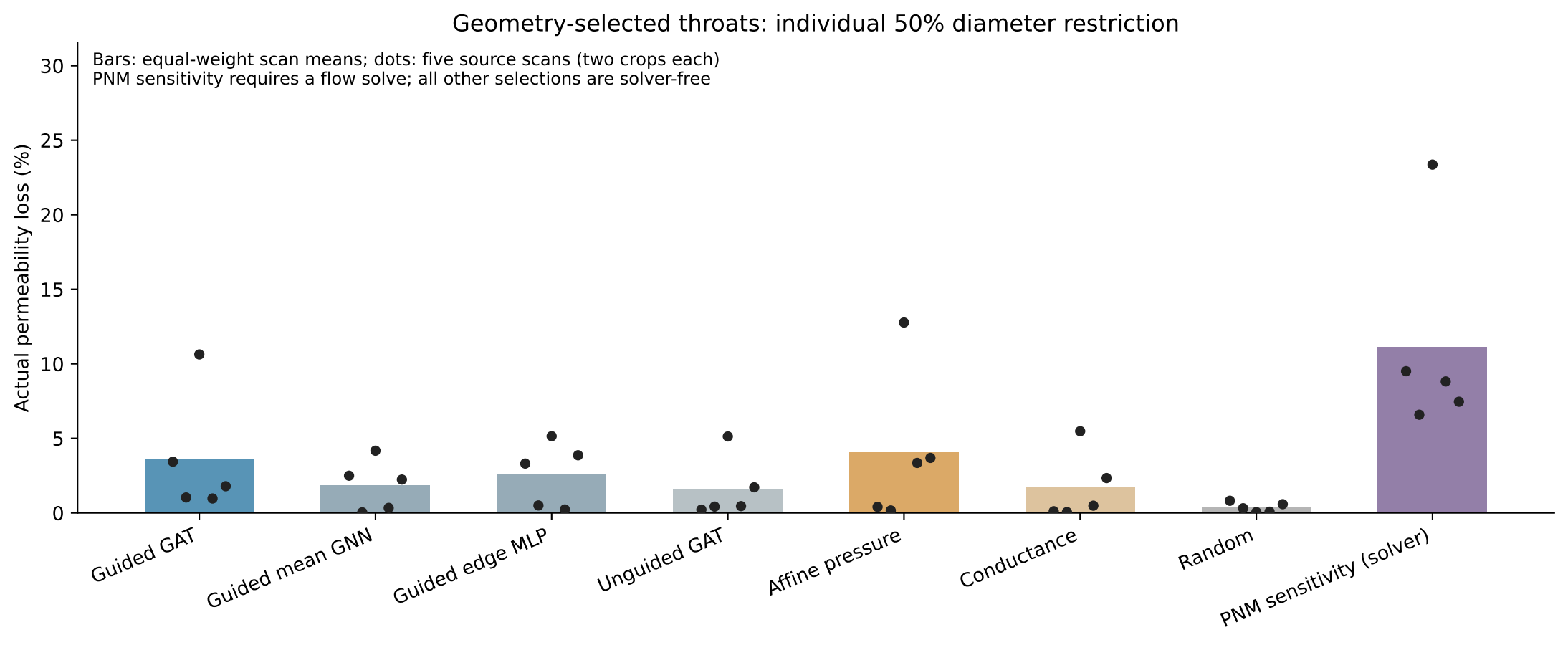
Geometry-selected throats: individual 50% diameter restriction
Bars: equal-weight scan means; dots: five source scans (two crops each)
PNM sensitivity requires a flow solve; all other selections are solver-free
Actual permeability loss (%)
0
5
10
15
20
25
30
Guided GAT
Guided mean GNN
Guided edge MLP
Unguided GAT
Affine pressure
Conductance
Random
PNM sensitivity (solver)